

Tyler Harnish

Manalapan, New Jersey

Tyler.M.Harnish@gmail.com | +1 (908) 202-4119 | [linkedin.com/in/tyler-harnish-th117](https://www.linkedin.com/in/tyler-harnish-th117) | github.com/TylerHarnish5

EDUCATION

Kean University — Union, NJ

Bachelor of Science in Computer Science, Magna Cum Laude | GPA 3.76 / 4.0

Relevant Coursework: Python Programming, Deep Learning & Applications, Web Application Development, Database Management Systems, Object-Oriented Analysis & Design, Computer Security, Human Computer Interaction

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, FastAPI, SQLAlchemy, PyTorch, torchvision, scikit-learn, Pandas, Tailwind CSS, Framer Motion

Databases & Backend: PostgreSQL, MySQL, REST APIs, Alembic, Pydantic

Tools & Platforms: Git, GitHub, Docker, Docker Compose, pytest, GitHub Actions, AWS EC2/RDS

EXPERIENCE

Technical Assistant

Fall 2024 – May 2026

Nancy Thompson Learning Commons, Kean University

- Supported and maintained a technology-focused lab (V.I.S.E.R), configuring projector display setups to meet client needs and providing hands-on technical assistance during presentations, university events, and Virtual Reality sessions.

General Laborer

June 2023 – Present

Unique Triangle Apparel, Marlboro, NJ

- Assisted with warehouse operations and inventory organization while supporting residents during on-site shopping events at assisted living facilities.

PROJECTS

Residential Clothing Automation Platform

Technologies: Python, FastAPI, AWS, PostgreSQL, SQLAlchemy, Alembic, Docker, pytest

- Built a Python/FastAPI order-management system supporting per-order resident budgets, inventory management, approval workflows, partial fulfillment, and persistent PostgreSQL storage.
- Designed REST APIs and relational data models with SQLAlchemy and Alembic, including persistent idempotency safeguards that prevent duplicate writes during request retries.
- Wrote 70+ pytest tests, configured GitHub Actions for automated testing and migration checks, and containerized the application with Docker Compose for local development and AWS EC2/RDS deployment.

Portfolio Web Application

Technologies: JavaScript, TypeScript, Next.js, React, PostCSS, Tailwind CSS, Framer Motion, Vercel

- Built a responsive personal portfolio using a component-based architecture to present projects, skills, and education.
- Developed animated and interactive UI components with Framer Motion and optimized layouts for desktop and mobile devices.
- Deployed the application to Vercel with continuous integration from GitHub, supporting rapid updates and version-controlled development.

TrashNet Waste Image Classification Using Deep Learning

Technologies: Python, PyTorch, torchvision, ResNet18, Pandas, Matplotlib, Hugging Face Datasets

- Trained a transfer-learning image classification model on 5,054 TrashNet images across six waste categories using a pretrained ResNet19 architecture.
- Evaluated architectural changes, dropout, batch normalization, and weight-decay regularization, achieving 95.65% test accuracy through iterative model experimentation.